WEEK 4 — GRAPHS & VISUALISATION CHEAT SHEET

All Sensors | matplotlib | FuncAnimation | Grade 8

1 SETUP

```
import matplotlib.pyplot as plt
import matplotlib.animation as anima

# Single graph
fig,ax=plt.subplots(figsize=(10,5))

# Two stacked graphs
fig,(ax1,ax2)=plt.subplots(2,1,figsi

# Three stacked
fig,(ax1,ax2,ax3)=plt.subplots(3,1)

# Title
fig.suptitle("Live Sensor",fontsize=
```

2 DRAWING

```
# ALWAYS clear first!
ax.clear()

# Basic line
ax.plot(times[-30:],vals[-30:])

# Styled line
ax.plot(times[-30:],vals[-30:],
        color=color,
        linewidth=2, marker="o")

# Fill under line
ax.fill_between(times[-30:],
    vals[-30:], alpha=0.2,
    color=color)
```

3 LABELS

```
ax.set_title("Temp: 28.5C")
ax.set_ylabel("Temperature C")
ax.set_xlabel("Time")
ax.set_ylim(0, 150)
ax.legend(loc="upper left")
plt.xticks(rotation=45, fontsize=7)
plt.tight_layout()
```

4 FUNCANIMATION

```
# Define animate function
def animate(i):
    # Read sensor
    line=ser.readline().decode(
        "utf-8",errors="ignore").str
    # Add to lists
    times.append(now)
    vals.append(value)
    # Clear and redraw
    ax.clear()
    ax.plot(times[-30:],vals[-30:],
            color=color)
    ax.set_ylabel("Value")
    plt.tight_layout()

# Start animation
ani=animation.FuncAnimation(
    fig, animate, interval=600)

plt.show()  # LAST LINE!
```

5 THRESHOLD LINES

```
# Danger line
ax.axhline(y=35, color="red",
           linestyle="--", linewidth=1,
           label="Hot 35C")

# Two levels
ax.axhline(y=60,color="red",linestyle="--")
ax.axhline(y=30,color="orange",linestyle="--")

ax.legend(loc="upper left",fontsize=8)
```

6 DYNAMIC COLORS

```
# Light sensor
color="gold" if lux>80 else "orange" if lux>30 else "gray"
# Temperature
color="red" if temp>35 else "orange" if temp>25 else "blue"
# Motion sensor
color="green" if score>0 else "red"
```

SENSOR Y-AXIS GUIDE

Light Sensor	0–150 lux
PIR Motion	0–110 score
DHT11 Temp	0–50 C
DHT11 Humidity	0–100 %
Sound Sensor	0–100 level
Rain Sensor	0–100 %
Flame Sensor	0–100 index
Vibration	0–20 IPS
Soil Moisture	0–100 %
Heart Rate	40–160 BPM

COMMON ERRORS

x and y same length?
Fix: Append times AND vals together

Graph blank/frozen
Fix: Never use while True + FuncAnimation

Labels overlapping
Fix: plt.xticks(rotation=45)+tight_layout()

Window closes instantly
Fix: plt.show() MUST be last line

GOLDEN RULES:

1. Upload Arduino code first, close IDE
2. plt.show() must always be the last line
3. ax.clear() before ax.plot() every frame!